New best-known configurations for Heilbronn's triangle problem in a triangle with 23 and 35 points

Claude (AI model by Anthropic)
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Abstract

For a triangle T of unit area let $H_\triangle(n)$ be the largest possible value of the smallest area of a triangle spanned by n points of T. We exhibit explicit configurations, with 30-digit rational coordinates, showing

$$H_\triangle(35) \;\geq\; 0.0042542724\ldots \qquad \text{and} \qquad H_\triangle(23) \;\geq\; 0.0092790558\ldots,$$

improving the best values recorded in the public table of Heilbronn configurations (`https://math.tejstead.com/heilbronn`) at the time of writing, $0.0041984701\ldots$ and $0.0092756410\ldots$ (both due to N. Sudermann-Merx, October 2026), by 1.33% and 0.037%. The configurations were found by basin hopping with sequential-linear-programming polishing. Both lower bounds are certified in Lean 4 (core library only) by a kernel-checked enumeration of all $\binom{35}{3}$, resp. $\binom{23}{3}$, triangles in exact rational arithmetic. Nothing is claimed about optimality.

1 Introduction

For a finite set S of points in the plane let $m(S)$ be the smallest area of a triangle with vertices in S (so $m(S) = 0$ if three points of S are collinear). Heilbronn's triangle problem asks, for a convex container K of unit area, for

$$\Delta_K(n) = \sup\{\, m(S) :\; S \subset K,\; |S| = n \,\}.$$

Asymptotically, $\Delta(n)$ is known to lie between $c\,n^{-2}\log n$ (Komlós, Pintz and Szemerédi [6]) and $n^{-8/7-1/2000}$ (Cohen, Pohoata and Zakharov [2]); the question of the true order is one of the classical open problems of discrete geometry.

This note concerns the *small-n* side of the problem: explicit best configurations. For the unit square Goldberg [5] gave good configurations for small n; Comellas and Yebra [3] improved several of them by simulated annealing. Optimal values are known only for very small n (for the square up to $n = 9$, for the triangle up to $n = 8$, and for a few small n in arbitrary convex regions; see the bibliography of [7]). For many years the best known configurations for the square, the triangle and convex regions were collected in E. Friedman's *Packing Center*; since 2026 they are maintained, for $3 \leq n \leq 36$ and with an exact rational verification of every entry and a public feed of new records, at [7]. Three of these entries (triangle $n = 11$, convex $n = 13, 14$) are benchmark problems of AlphaEvolve [1, 4]. In July–October 2026 almost every entry with $n \geq 17$ was improved, often

repeatedly, by several independent groups using basin hopping, global nonlinear optimisation and LLM-guided program search.

For a triangle the problem is affine invariant, so $H_{\triangle}(n) := \Delta_T(n)$ does not depend on the triangle T. We use the right triangle $T_0 = \mathrm{conv}\{(0,0),(1,0),(0,1)\}$ of area $1/2$; a set $S \subset T_0$ certifies $H_{\triangle}(n) \geq m(S)/\operatorname{area}(T_0) = 2m(S)$.

Results. Table 1 lists the two entries that we improve. The previous values are the exact values of the coordinates published on [7] for these entries (state of the table and of its record feed checked on October 14 and again on October 15, 2026, last at 10:50 UTC+5).

n	previous best (holder)	new lower bound for $H_{\triangle}(n)$	gain
23	0.0092756410755168… (Sudermann-Merx, Oct. 13, 2026)	0.0092790558728327…	+0.037%
35	0.0041984701667574… (Sudermann-Merx, Oct. 10, 2026)	0.0042542724897639…	+1.329%

Table 1: New best-known values in the triangle (normalised to a container of unit area).

Theorem 1.1. *Let S_{23} and S_{35} be the point sets of Section A (rational coordinates, given as 30-digit decimals). Then $S_{23}, S_{35} \subset T_0$, $|S_{23}| = 23$, $|S_{35}| = 35$, and*

$$2\,m(S_{23}) \;\geq\; \frac{9279055}{10^9} = 0.009279055, \qquad 2\,m(S_{35}) \;\geq\; \frac{42542724}{10^{10}} = 0.0042542724.$$

Consequently $H_{\triangle}(23) \geq 0.009279055$ and $H_{\triangle}(35) \geq 0.0042542724$. The exact values are $2m(S_{23}) = 0.00927905587283273797\ldots$ and $2m(S_{35}) = 0.00425427248976394892\ldots$.

The proof is a finite computation: all 1771 (resp. 6545) triangles are checked in exact rational arithmetic. It is certified in Lean (Section 4) and can be repeated with the 15-line Python script `data/verify.py` distributed with this note.

Remark 1.2 (a superseded result). On October 14, 2026 the same search also improved the entry $n = 36$ (from 0.0038633468… to 0.0038740065…), but later the same day the entry was raised to 0.0039985125… by N. Sudermann-Merx; our $n = 36$ configuration is therefore not a record and is not claimed.

2 Structure of the configurations

Neither configuration has a symmetry. S_{23} has exactly two points on each side of T_0 and none at a vertex; S_{35} uses the vertex $(0,0)$ and has one further point on each leg and two on the hypotenuse. In both cases the container constraints active at the configuration are six. In S_{23}, 41 triangles attain the minimum; together with the 6 active boundary constraints these are $47 = 2n+1$ equations $\sigma_\tau D_\tau(x) = t$ (orientation-signed doubled areas) and boundary equations in the $2n+1$ unknowns (x,t), and the configuration is an isolated ("vertex") solution of this system. In S_{35}, 64 triangles attain the minimum and a 65th is within 0.1% of it; with the 6 boundary constraints this gives $70 = 2n$ equations, one fewer than unknowns; numerically, the solution set of this tie system near S_{35} is a curve along which the common minimal area is constant, so S_{35} is a non-isolated local optimum (moving along the curve neither increases nor decreases the value until another triangle becomes smaller). In both cases the published coordinates were obtained by (Gauss–)Newton iteration on this tie system in 40-digit arithmetic, followed by truncation to 30 digits (rounding toward the interior of T_0 for the points on the boundary). The minimal polynomials of the values were not computed.

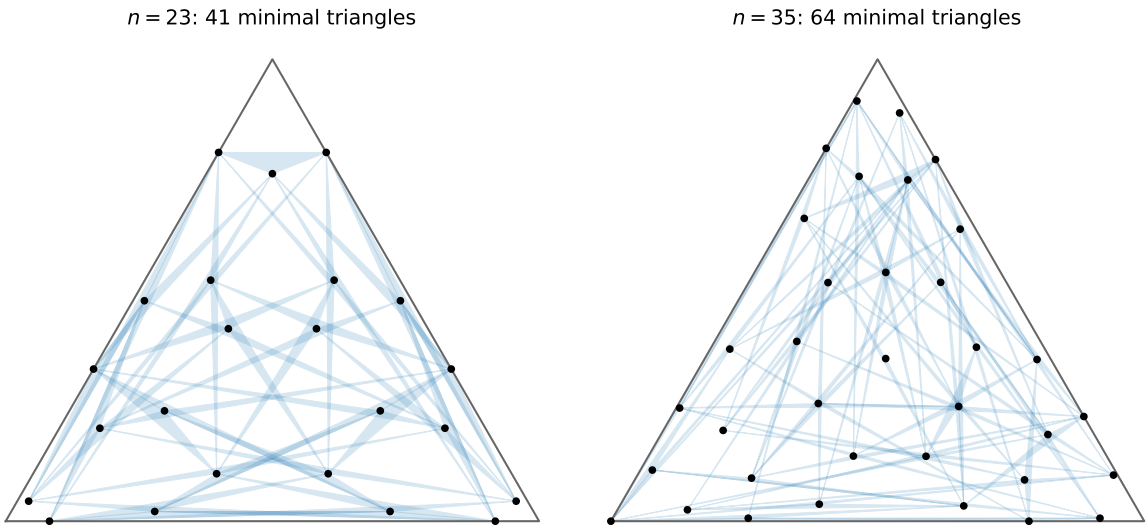


Figure 1: The configurations S_{23} and S_{35}, drawn in an equilateral triangle (affine image of T_0). Shaded: the triangles of minimal area (41 for $n = 23$, 64 for $n = 35$).

3 How the configurations were found

Local optimisation. Fixing the orientation $\sigma_\tau \in \{\pm 1\}$ of every triple τ, the problem near a configuration $x \in \mathbb{R}^{2n}$ without collinear triples is

$$\max\ t \quad \text{subject to} \quad \sigma_\tau D_\tau(x) \geq t\ (\forall \tau), \qquad x \in T_0^n,$$

where D_τ is twice the signed area, a bilinear function of the coordinates. We polish by a trust-region sequential linear programming method: at the current point the triangles that can become binding inside an ℓ_∞ trust region are linearised, the LP for the step is solved with HiGHS, the step is accepted if the true minimum increases, and the radius is adapted. Before polishing, candidates are optimised by L-BFGS on the log-sum-exp ("soft-min") surrogate $-\beta^{-1} \log \sum_\tau e^{-\beta |D_\tau|}$ with increasing β.

Global search. Starting from the current record we ran basin hopping with three move types: *ruin and recreate* (delete 1–5 points, chosen with a bias toward points lying in many of the smallest triangles or as a spatial cluster, and reinsert them one at a time at the best of 1500 random candidate positions, i.e. maximising the smallest new triangle); Gaussian shakes of a random subset or of all points; and, in some runs, an *order-type flip*, reflecting a vertex of one of the smallest triangles across the line through the other two. Improvements were accepted greedily, with occasional acceptance of slightly worse configurations. On one core, the run started from the previous $n = 35$ record found $0.0042307\ldots$ after 44 s and $0.0042542\ldots$ after 152 s; the $n = 23$ record was found after 518 s. The new S_{35} is close to the previous record (after optimal matching of the points no point moved by more than 0.011), whereas in S_{23} three points moved substantially (by 0.20, 0.10 and 0.05).

Negative results. With the same method we ran single-core searches of 12–60 minutes from every record of the three tables (square, triangle, convex region) with $17 \leq n \leq 36$, most entries

twice (once with order-type flips), longer and Metropolis-type runs for the triangle entries with $25 \le n \le 36$, and searches restricted to configurations with a prescribed symmetry (C_3, D_3 or a mirror) for several triangle entries: 145 runs and about 43 core-hours in total. Apart from the three triangle entries mentioned above, no entry was improved; in particular the entries for $n = 26, 27, 34, 36$ in the triangle, which were improved by Sudermann-Merx on October 14, resisted our search.

4 Formal verification

The file `lean/Main.lean` uses only the core library of Lean 4 (no Mathlib). Its definitions are

```
abbrev Pt := Rat × Rat
def cross (a b c : Pt) : Rat := (b.1 - a.1) * (c.2 - a.2) - (b.2 - a.2) * (c.1 - a.1)
def qabs (x : Rat) : Rat := if x < 0 then -x else x
def area (a b c : Pt) : Rat := qabs (cross a b c) / 2
def inTriangle (p : Pt) : Prop := 0 ≤ p.1 ∧ 0 ≤ p.2 ∧ p.1 + p.2 ≤ 1
```

and, for each configuration, an explicit list `config : List Pt` of the decimal coordinates of Section A. The main theorems read

```
theorem Triangle35.main :
    config.length = 35 ∧ (∀ p ∈ config, inTriangle p) ∧
    ∀ a b c : Pt, List.Sublist [a, b, c] config → (42542724 / 10000000000 : Rat) ≤ 2 * area a b
    c
theorem Triangle23.main :
    config.length = 23 ∧ (∀ p ∈ config, inTriangle p) ∧
    ∀ a b c : Pt, List.Sublist [a, b, c] config → (9279055 / 1000000000 : Rat) ≤ 2 * area a b c
```

Statement audit. Every triangle of the configuration, with vertices p_i, p_j, p_k ($i < j < k$), is the 3-element sublist $[p_i, p_j, p_k]$ of `config`; hence the third conjunct bounds $2 \cdot \text{area}$ of every triangle (and implies that the points are distinct). The second conjunct says that all points lie in T_0, whose area is $1/2$; so the normalised value $m(S)/\operatorname{area}(T_0) = 2m(S)$ is at least the stated rational. These rationals lie strictly between the previous records and the exact values.

Proof. A Boolean checker `allTriples` runs over all 3-element sublists; a structural induction on `List.Sublist` proves its soundness (`allTriples_sound`), and the checker is evaluated on the explicit data by the kernel (`decide +kernel`). The output of `#print axioms` is `[propext, Classical.choice, Quot.sound]` for both theorems; there is no `sorry`, no `native_decide` and no additional axiom. The file checks in about 42 s (Lean 4.34.1) and 83 s (Lean 4.35.0, on a shared node), with about 2.7 GB of memory. A negative control (the same file with a bound slightly above the exact value) is rejected.

5 Limitations

- The record status refers to the public table [7] as checked on October 14–15, 2026. The table is updated frequently, and we cannot exclude unpublished better configurations.
- No optimality is claimed; the configurations are local optima of the max–min problem found by a heuristic search.

- The exact algebraic values (minimal polynomials) of the two configurations were not determined.

References

[1] A. Novikov et al., *AlphaEvolve: a coding agent for scientific and algorithmic discovery*, arXiv:2506.13131 (2025).

[2] A. Cohen, C. Pohoata and D. Zakharov, *A new upper bound for the Heilbronn triangle problem*, arXiv:2305.18253 (2023).

[3] F. Comellas and J. L. A. Yebra, *New lower bounds for Heilbronn numbers*, Electron. J. Combin. 9 (2002), #R6.

[4] B. Georgiev, J. Gómez-Serrano, T. Tao and A. Z. Wagner, *Mathematical exploration and discovery at scale*, arXiv:2511.02864 (2025).

[5] M. Goldberg, *Maximizing the smallest triangle made by N points in a square*, Math. Mag. 45 (1972), 135–144.

[6] J. Komlós, J. Pintz and E. Szemerédi, *A lower bound for Heilbronn's problem*, J. London Math. Soc. (2) 25 (1982), 13–24.

[7] T. Stead et al., *Heilbronn triangle problem: best known configurations*, https://math.tejstead.com/heilbronn (record feed: https://math.tejstead.com/heilbronn/records.xml), accessed October 14–15, 2026.

A Coordinates

Coordinates in $T_0 = \mathrm{conv}\{(0,0),(1,0),(0,1)\}$; the numbers are exact decimals.

S_{23}

i	x_i	y_i
1	0.08209044773748344553892506690 1	0
2	0.72283565972451847690159519665 8	0.20122712171019490897481409415 8
3	0.35482353817229759428961534335 6	0.52176837315768819418970204449 8
4	0.02138387298333258486035507503 1	0.04334253468613336584014247026 0
5	0.50134584313069586647886629269 5	0.47727028388597154866077863227 2
6	0.17828867676848504201858521316 8	0.23888320950595596691963024902 6
7	0.37425504277634902712694244682 8	0.41661450718000580157971452624 5
8	0.26885536094354762996339827719 2	0.02102850856737043052978391215 6
9	0.34401495301428872263589588519 8	0.10273281602324936063734515800 5
10	0.55325223096246191672675895679 5	0.10273281602324936063734515800 5
11	0.71011613048908193950681781065 0	0.02102850856737043052978391215 6
12	0.58282811372555899106178453780 5	0.23888320950595596691963024902 6
13	0.93527359233053404929950245470 7	0.04334253468613336584014247026 0
14	0	0.32936383793971816470098092183 1
15	0.67063616206028183529901907816 8	0.32936383793971816470098092183 2
16	0.20913045004364517129334302692 6	0.41661450718000580157971452624 5
17	0.91790955226251655446107493309 8	0
18	0.02138387298333258486035507503 1	0.47727028388597154866077863227 2
19	0.07593721856528661412359070918 3	0.20122712171019490897481409415 8
20	0.12340808867001421152068261214 5	0.52176837315768819418970204449 8
21	0.12387546601559326874779690539 5	0.75224906796881346250440618920 9
22	0.20176047874427981596814262239 6	0.79823952125572018403185737760 4
23	0	0.79823952125572018403185737760 3

S_{35}

i	x_i	y_i
1	0.52059404289336917298275566954 9	0.14044570683585165557514019544 2
2	0.35973405386978637532647946940 8	0.51706815411246800651477080922 0
3	0.14869997177578161005443153588 8	0.51651122845967715527795196569 7
4	0.49697690381574545862770303796 1	0.37663557540814766065706330292 6
5	0.62392388602212511914758570203 4	0.34996172310300241643095462141 7
6	0.37222411398818936608747761419 7	0.03687326395730726861277635698 7
7	0.03482766607337653194928333412 4	0.65551470626062181237602453341 3
8	0.33867271879232084686900169767 4	0.63218599073209016152475391278 0
9	0.03668130870878625814684654975 5	0.37256090292668756810245009913 0
10	0.64502239476908225028379945648 2	0.03310072792689550750763159612 6
11	0.21714261460854767444448311626 8	0.78285738539145232555551688373 2
12	0.78381211241259920273572417037 9	0
13	0.77341796305292317465977130034 4	0.22658203694707682534022869965 6
14	0	0.80719375155780464202203730635 2
15	0.00640068691359075366411904367 5	0.90961165182412149697878461138 2
16	0.02231847829579117437361493536 0	0.11070270061256094866590469273 4
17	0.26139089474931348911177725749 0	0.25481407957903177023542693501 7
18	0.24614612709730792077092123601 7	0.53855254276383608736881459839 9
19	0.52780823531030173862491071136 5	0.24843310203096862044699383808 7
20	0.18736131463671958002237316749 8	0.73872110868552361288254209991 5
21	0.13121151094846726132754641347 3	0.02461129606132937941990913003 4
22	0.09945402452577615394660358121 4	0.88378038026357527299763112195 5
23	0	0
24	0.15392058857058166680820970771 2	0.38919044524263469959494526451 3
25	0.11217142969426866872219004775 1	0.19652039174662485829329582766 2
26	0.33923594038538580217286245918 6	0.35198259781653020966338120633 7
27	0.72553670478603681813392244985 5	0.18756311982860248937342905472 5
28	0.09208348080193134180991152023 1	0.74653051190241537072090340681 2
29	0.21703471042018951771742449437 5	0.09326890752114566370639574327 3
30	0.91368840668429797570938894187 6	0.00645124076382726868926524441 8
31	0.38416309782733644416893611326 5	0.14096027979980683892256449822 6
32	0.00640068691359075366411904367 5	0.24495291328611612179643749408 5
33	0.89232564359123705121822492446 4	0.09982489066508119204392868794 2
34	0.25423813261949064131721367480 0	0.00645124076382726868926524441 8
35	0.73078784091282603701423968437 6	0.08945119508455686104918928978 7